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Q No. 1

The position function is $s(t) = 7t^3 - 3t^2 - 5t + 2$.

- Instantaneous velocity $v(t)$ is the first derivative of $s(t)$:

$$v(t) = \frac{ds}{dt} = 21t^2 - 6t - 5$$

At $t = 5$:

$$v(5) = 21(5)^2 - 6(5) - 5 = 21(25) - 30 - 5 = 525 - 35 = 490.$$

- Acceleration $a(t)$ is the second derivative of $s(t)$:

$$a(t) = \frac{d^2s}{dt^2} = 42t - 6$$

At $t = 5$:

$$a(5) = 42(5) - 6 = 210 - 6 = 204.$$

Answer: Instantaneous velocity at $t = 5$ is 490 , and acceleration is 204

Q No. 2

The function is $f(x, y) = \frac{x^2y}{x^2+y^2}$.

To evaluate $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$, use polar coordinates:

Let $x = r \cos \theta$, $y = r \sin \theta$.

$$\text{Then } f(x, y) = \frac{(r \cos \theta)^2 (r \sin \theta)}{(r \cos \theta)^2 + (r \sin \theta)^2} = \frac{r^3 \cos^2 \theta \sin \theta}{r^2 (\cos^2 \theta + \sin^2 \theta)} = r \cos^2 \theta \sin \theta.$$

As $(x, y) \rightarrow (0,0)$, $r \rightarrow 0$, and $|f(x, y)| = |r \cos^2 \theta \sin \theta| \leq |r| \rightarrow 0$.

By the squeeze theorem, the limit is 0 .

Answer: The limit is 0 .

Q No. 3

The temperature function is $T(x, y) = 4x^2y - y^3 + 2$.

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- Partial derivative with respect to x :

$$\frac{\partial T}{\partial x} = \frac{\partial}{\partial x} (4x^2y - y^3 + 2) = 8xy$$

At (1,2) :

$$\frac{\partial T}{\partial x} = 8(1)(2) = 16$$

- Partial derivative with respect to y :

$$\frac{\partial T}{\partial y} = \frac{\partial}{\partial y} (4x^2y - y^3 + 2) = 4x^2 - 3y^2$$

At (1,2) :

$$\frac{\partial T}{\partial y} = 4(1)^2 - 3(2)^2 = 4 - 12 = -8$$

Answer: $\frac{\partial T}{\partial x} = 16$ and $\frac{\partial T}{\partial y} = -8$ at (1,2).

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